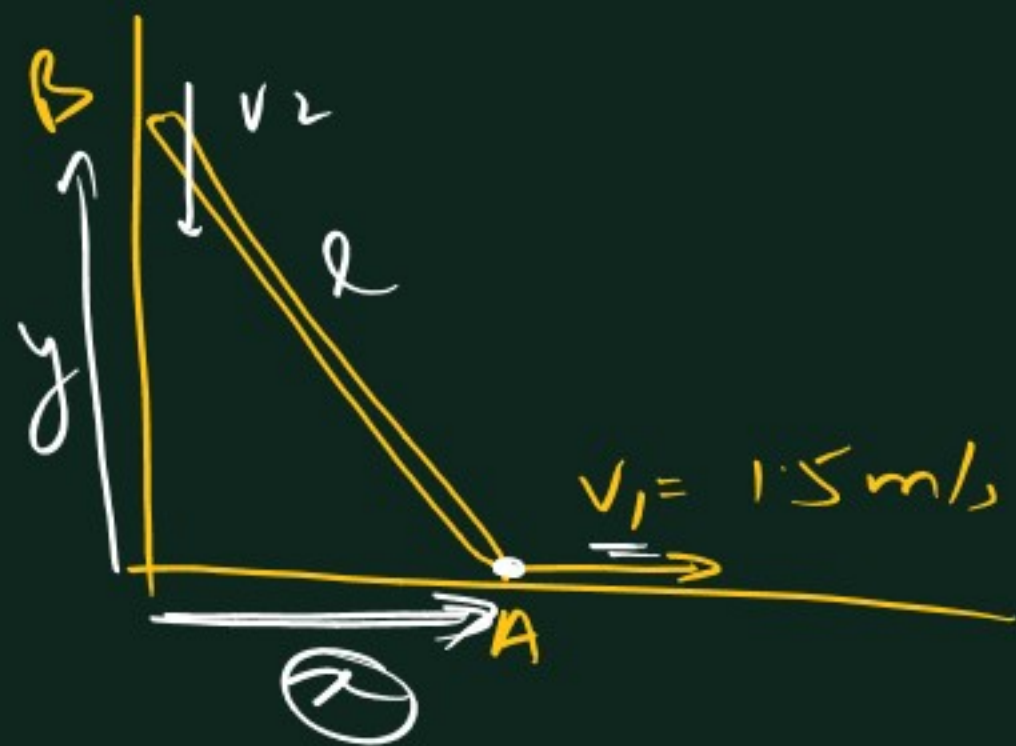


26 $\int \sin 2x \, dx + \int \cos 2x \, dx$
 $= \cancel{x} \left(- \cancel{\frac{1}{x}} \frac{|\cos 2x|}{\cancel{x}} + \frac{|\sin 2x|}{\cancel{x}} \right)$

19)



$$\rightarrow x^2 + y^2 = 25 - \textcircled{1}$$

$$\rightarrow \underline{x} \uparrow \Rightarrow \frac{dx}{dt} = v_1 \text{ or } 1.5 - \textcircled{2}$$

$$\text{To find } v_2 = \frac{dy}{dt}$$

Differentiate ①

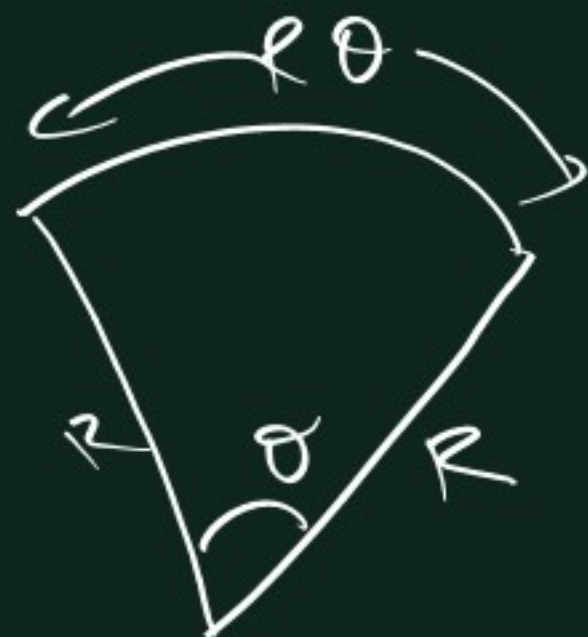
$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$$

$$\Rightarrow \frac{dy}{dt} = - \frac{x \frac{dx}{dt}}{y}$$

$$= - \frac{4}{3} \times 1.5$$

$$= \textcircled{-2}$$

23



$$P = 2R + R\theta \quad \text{--- (1)}$$

$$\Rightarrow \theta = \frac{P - 2R}{R}$$

$$\underline{\underline{\text{Area}}} = \cancel{\pi R^2} \times \frac{\theta}{\cancel{2\pi}} = \underline{\underline{\frac{R^2 \theta}{2}}} \quad \text{--- (2)}$$

$$A = \frac{R^2 (P - 2R)}{2}$$

$$A = \frac{R^2 P}{2} - R^2$$

$$\frac{dA}{dR} = 0$$

$$\Rightarrow \frac{P}{2} \times 1 - 2R = 0$$

$$\Rightarrow \underline{\underline{R = \frac{P}{4}}}$$

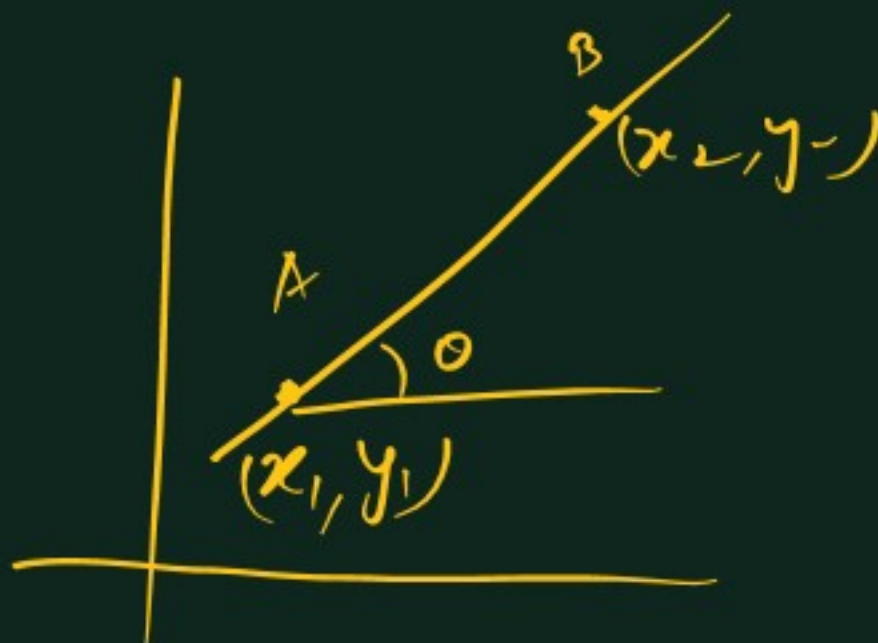
Graphs

1) $y = mx + c$

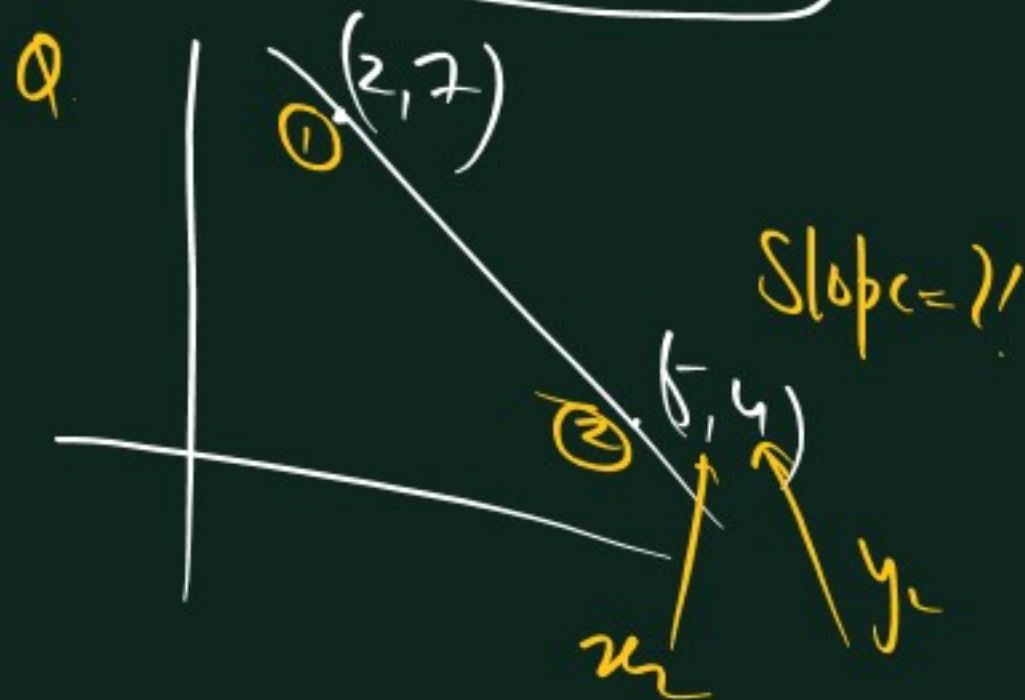
2) $a > 0$

$a < 0$

$y = ax^2 + bx + c$



$$\text{Slope} = \frac{y_2 - y_1}{x_2 - x_1}$$

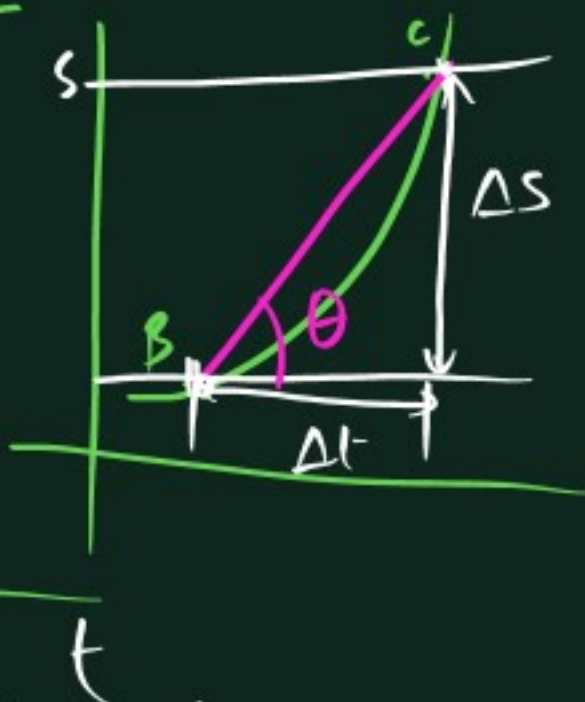


$$\begin{aligned} \text{Slope} &= \frac{4 - 7}{5 - 2} \\ &= -\frac{3}{3} \\ &= -1 \end{aligned}$$

	Slope	Area	More info
1) <u>$x-t$</u>	$ \begin{array}{c} v \\ \swarrow \quad \searrow \\ \text{inst} \quad \text{Avg.} \end{array} $	⊗	
2) <u>$v-t$</u>	$ \begin{array}{c} a \\ \swarrow \quad \searrow \\ \text{inst.} \quad \text{av.} \end{array} $	Disp. (Distance)	→ Avg. speed → Avg. velocity → inst. speed
3) $a-t$	⊗	Change in velocity	

$$\left(\frac{dv}{dt}\right), \int v dt$$

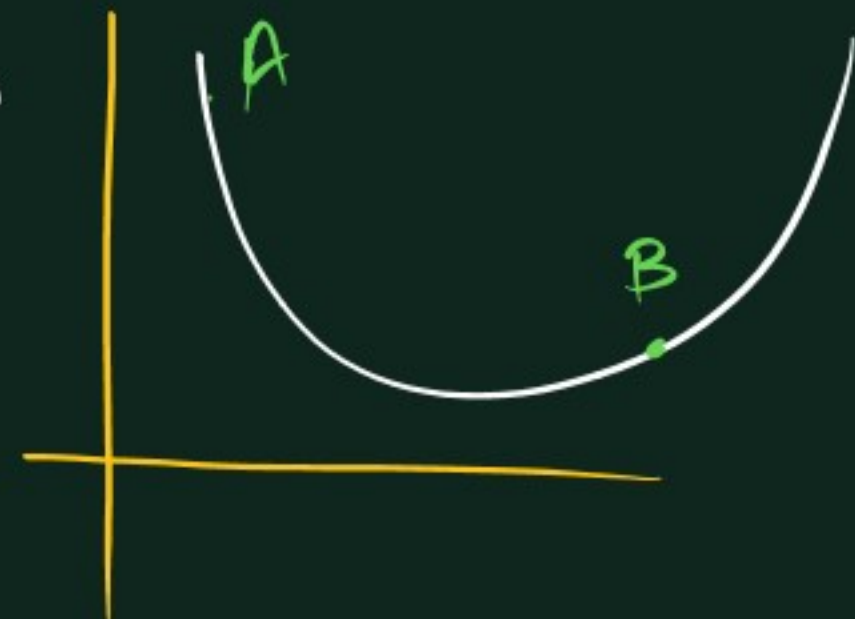
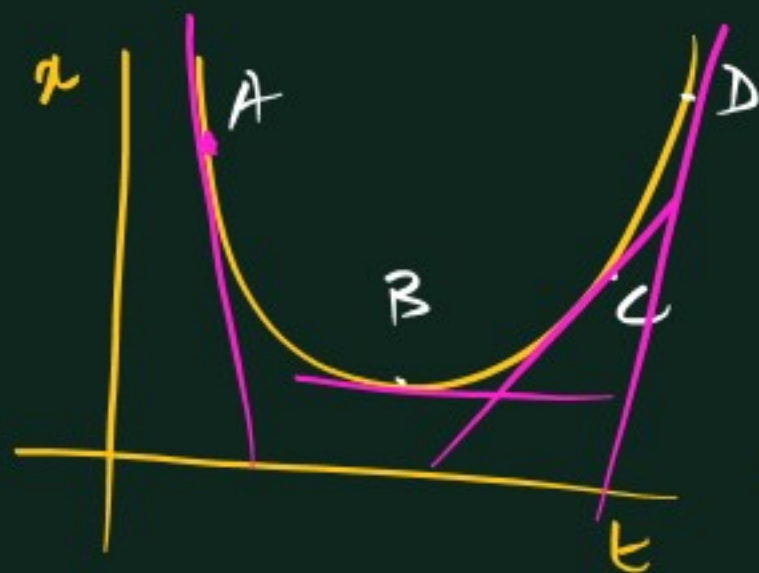
Position time / Disp-time.



→ $\tan \theta = v$ at A (instantaneous velocity)

→ B to C Avg. velocity = $\frac{\text{Displacement}}{\text{time}}$
 $= \left(\frac{\Delta s}{\Delta t} \right)$

Q1.



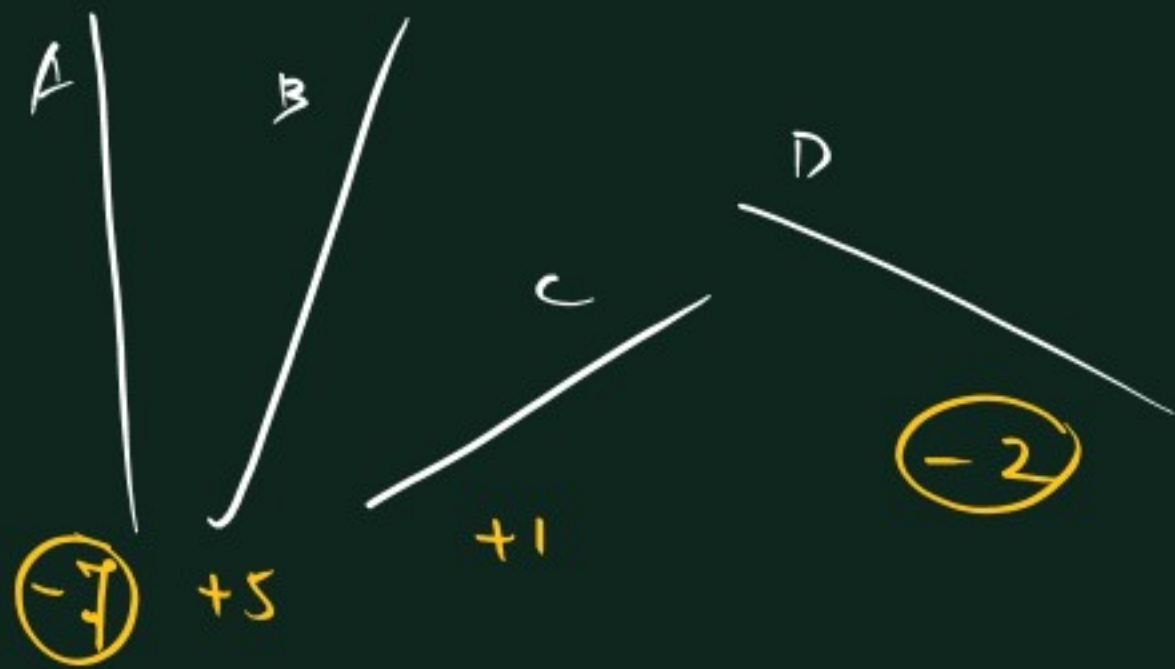
1) At which point particle is moving towards left

2) Compare speeds at B, C, D.

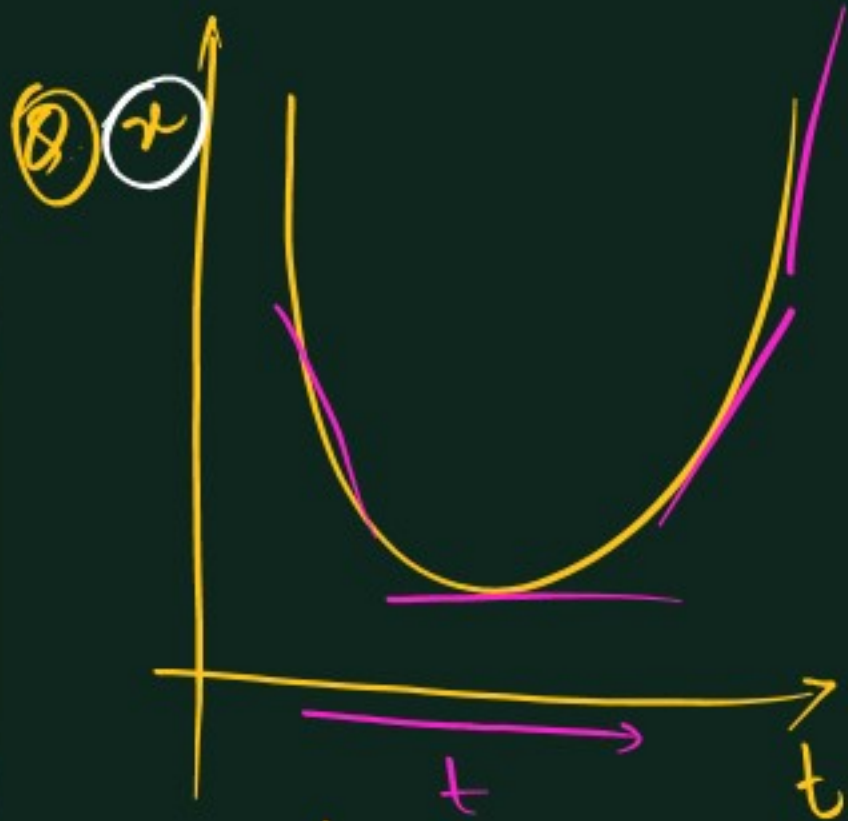
1) A → as $v \ominus$ at A
 (slope $\ominus$)



2) $v_B < v_C < v_D$ (Steeper $\Rightarrow$ velocity/slope more)



Slopes $\rightarrow -2, -7, +1, +5$



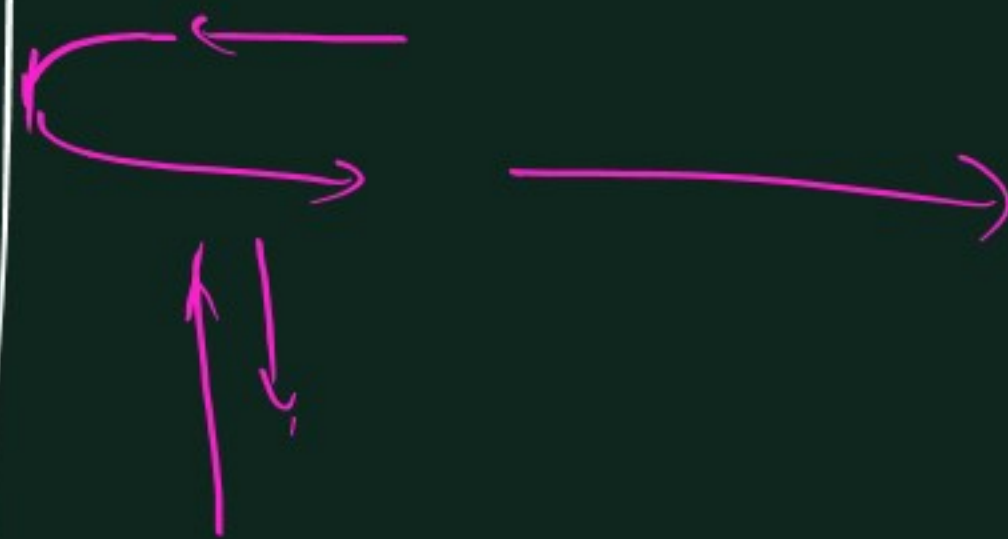
If particle is moving with constant 'a', is the 'a' positive or negative.
 $(\rightarrow)$ $(\leftarrow)$

$$1) \quad x = ut + \frac{1}{2} at^2$$

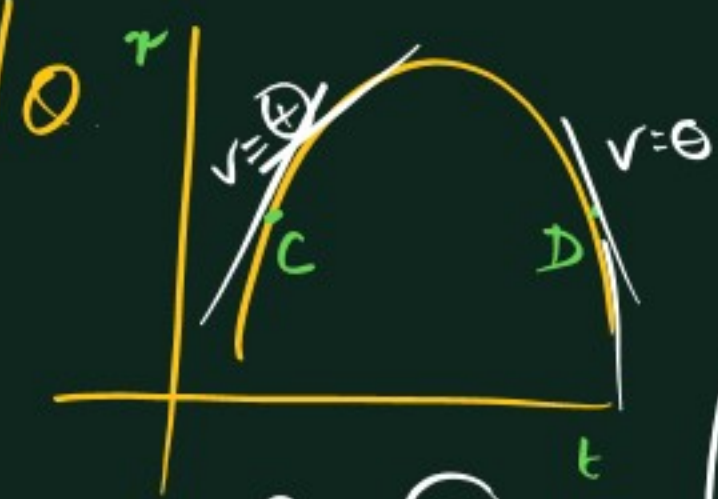
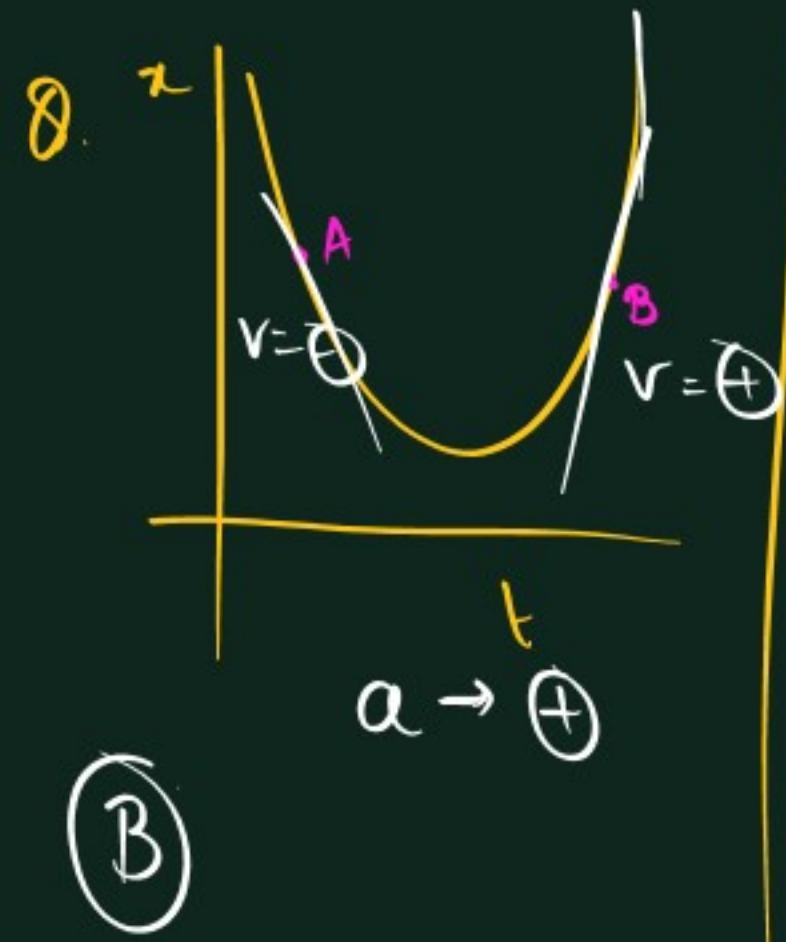
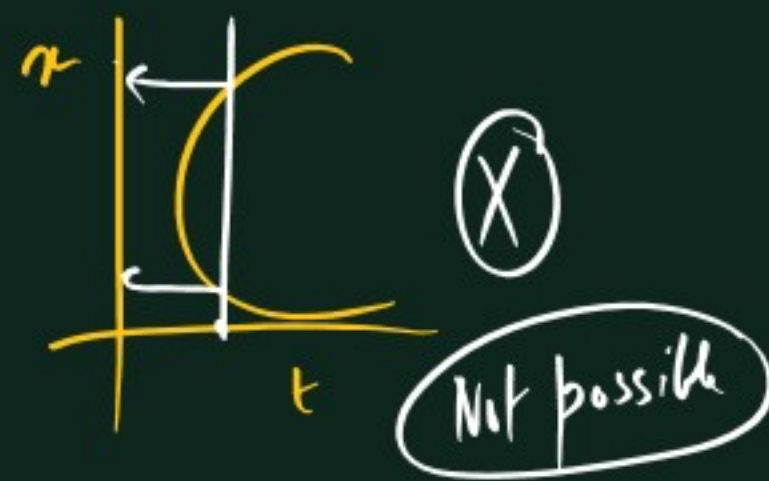
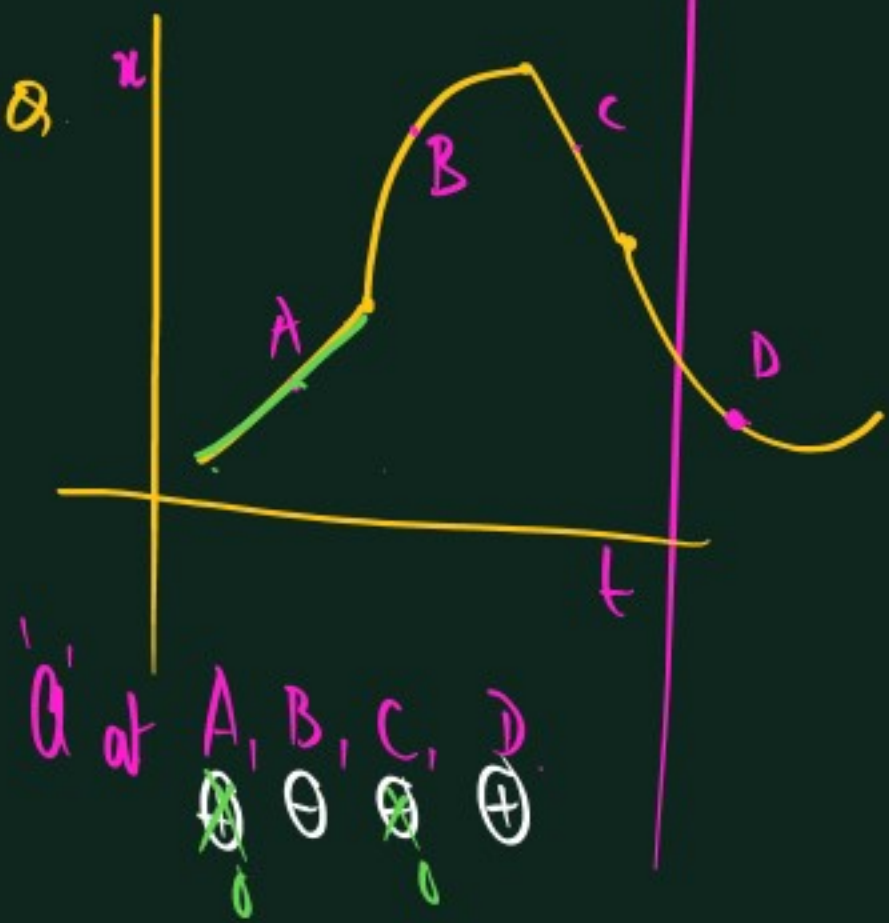
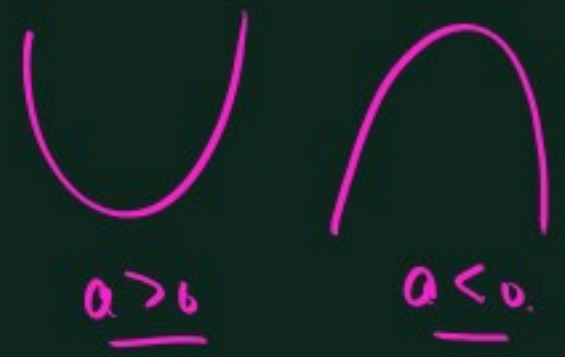
$$x = \left(\frac{a}{2}\right)t^2 + ut$$

if $\frac{a}{2} > 0$
 or $a > 0$

(or)



(2k)



$a = -$

A B C D

Speed ↑

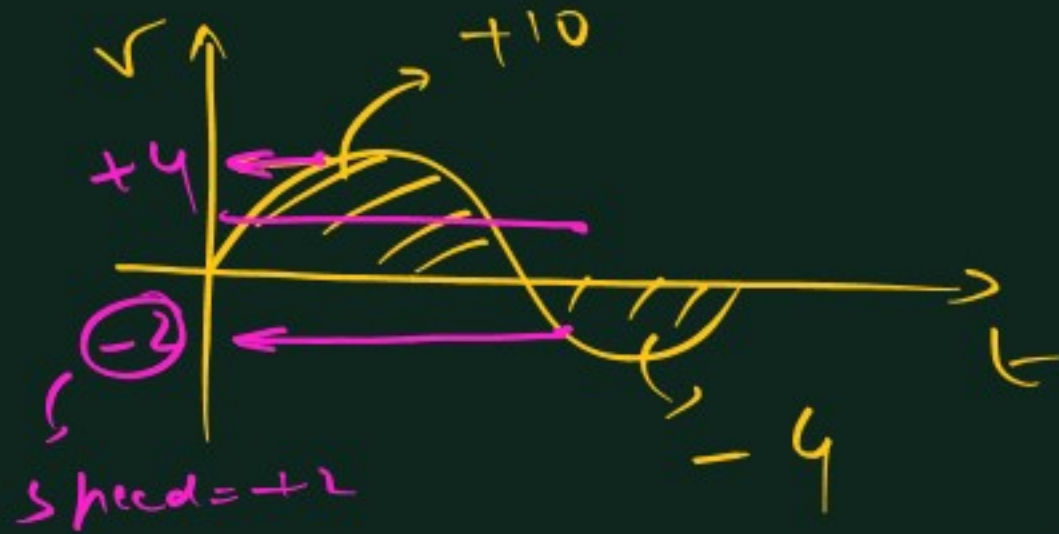


	v	a
A →	-	+
B	+	+
C	+	-
D	-	-

accelerated

$$\frac{\Delta v}{\Delta t}$$

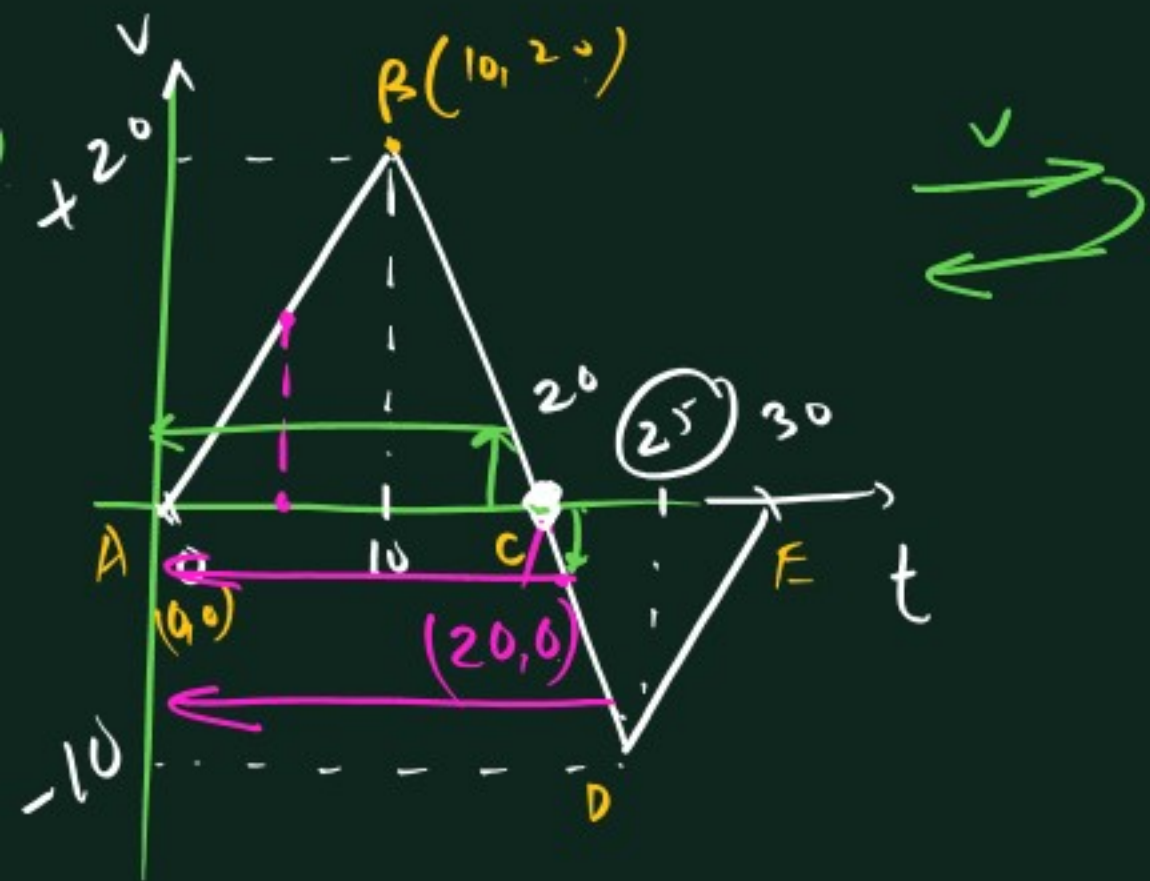
$$\left(\int v dt \right)$$



$$\text{Displacement} = 10 + (-4) = \underline{6\text{ m}}$$

$$\text{Distance} = 10 + 4 = \underline{14\text{ m}}$$





1) a at $5s = \text{slope of } AB = \frac{y_2 - y_1}{x_2 - x_1} = \frac{20 - 0}{10 - 0} = \underline{2 \text{ m/s}^2}$

$a = 5s = \text{slope of } BC/CD/BD = \frac{y_2 - y_1}{x_2 - x_1} = \frac{0 - 20}{20 - 10} = \underline{-2 \text{ m/s}^2}$

2) $\text{avg } a = \frac{\text{change in } v}{\text{time}} = \frac{v_2 - v_1}{t} = \frac{(-10) - 0}{25} = \underline{-0.4 \text{ m/s}^2}$

3) $\frac{1}{2} \times 20 \times 20 = 200$
 $\frac{1}{2} \times 10 \times 10 = -50$

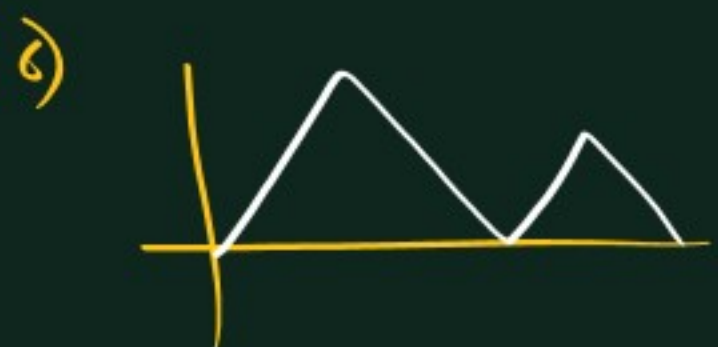
Disp. = $200 - 50 = 150 \text{ m}$
 Distance = $200 + 50 = 250 \text{ m}$

4) Avg. velocity = $\frac{\text{Disp.}}{t} = \frac{150}{30} = 5 \text{ m/s}$

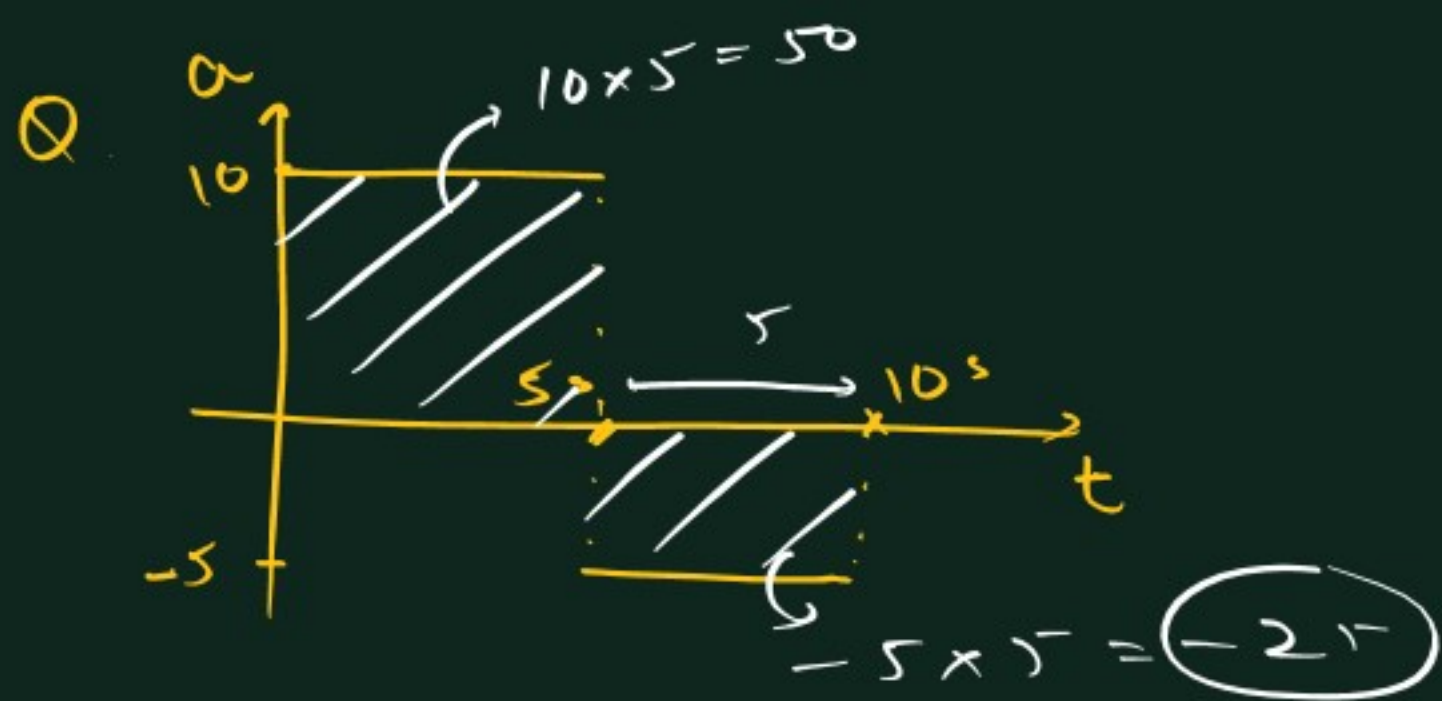
Avg. speed = $\frac{\text{Distance}}{\text{time}} = \frac{250}{30} = \underline{25/3 \text{ m/s}}$

5) $\rightleftharpoons$
 $\rightleftharpoons$
 Turn $\Rightarrow v$ changes sign

$t = 20s$
 $\oplus$
 $\ominus$
 20s



	v	a
AB	+	+
BC	+	-
CD	-	-
DE	-	+

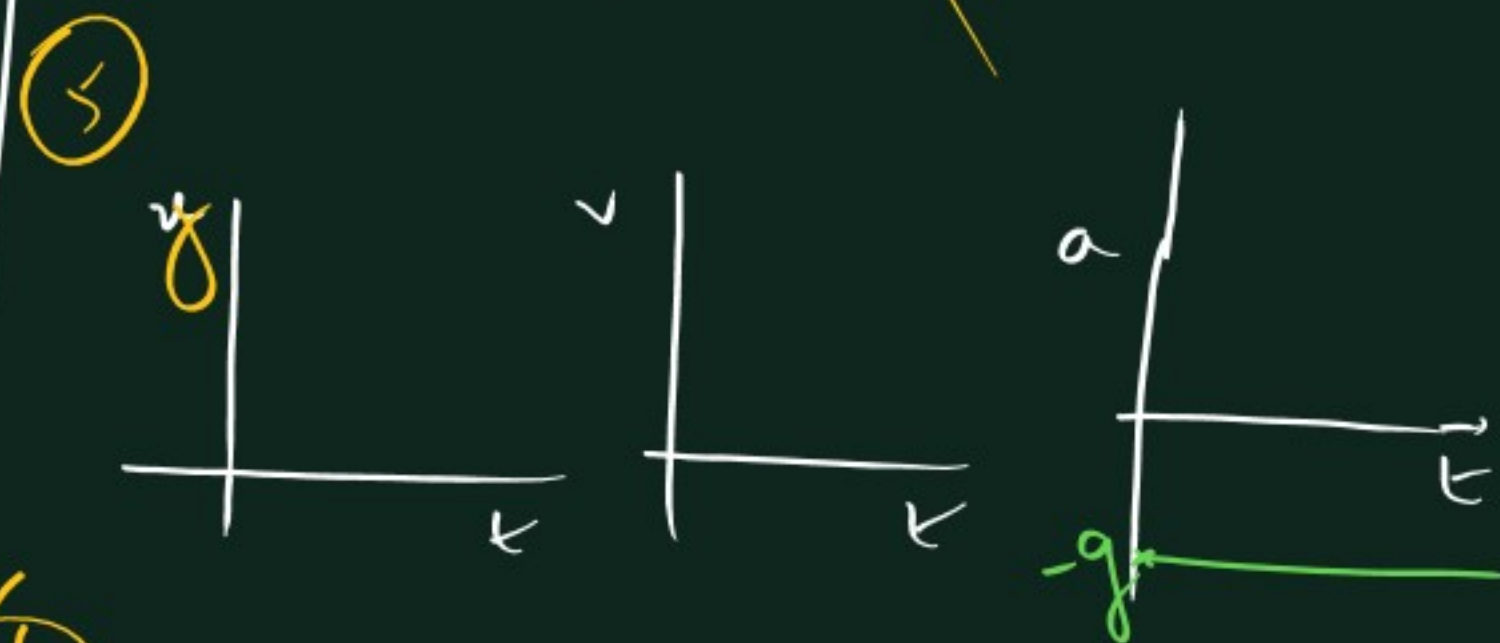
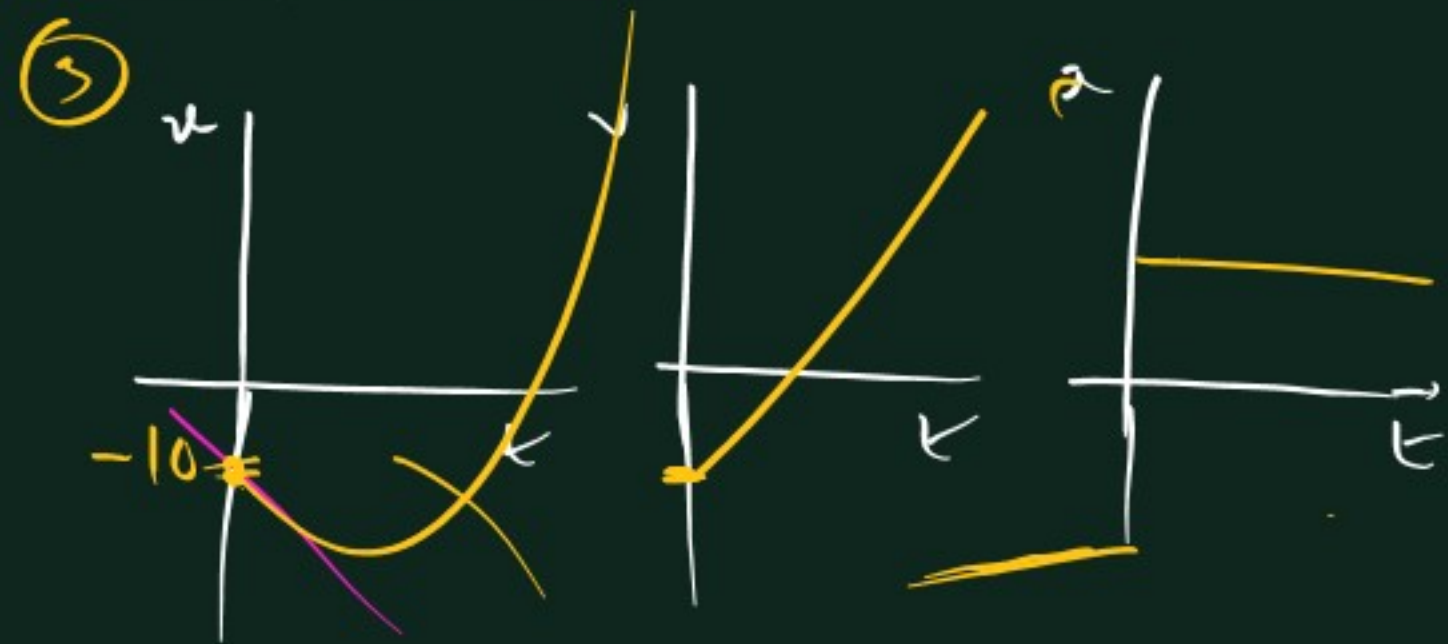
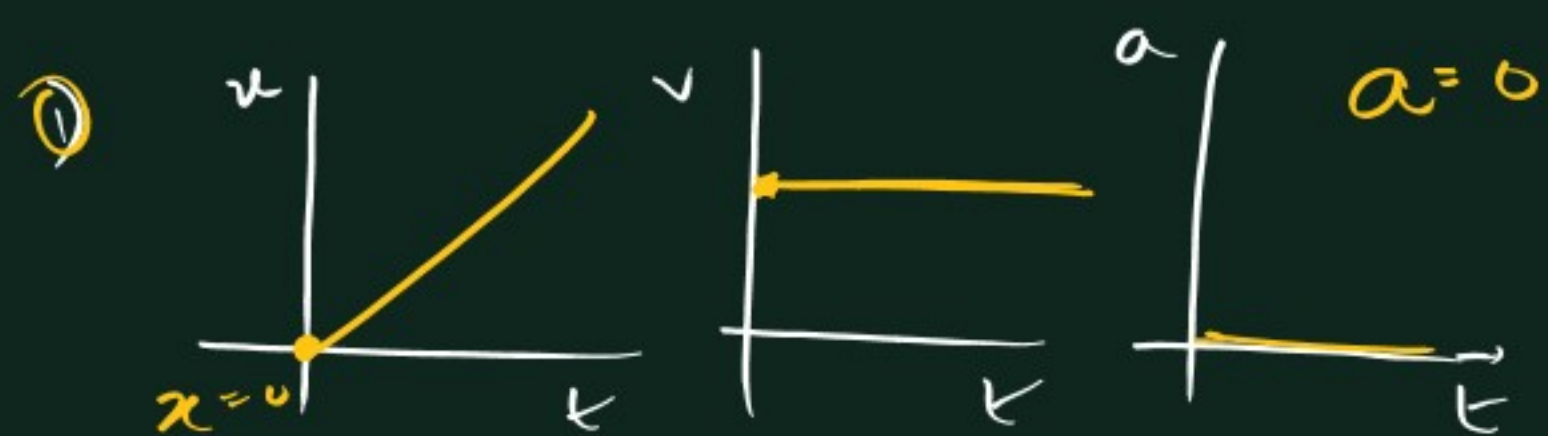


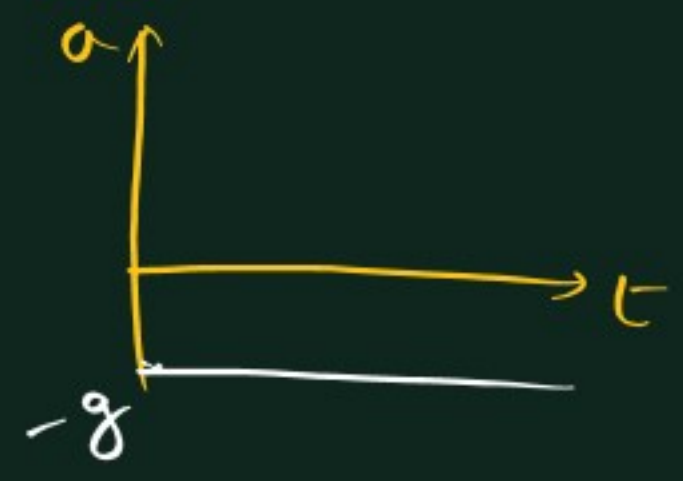
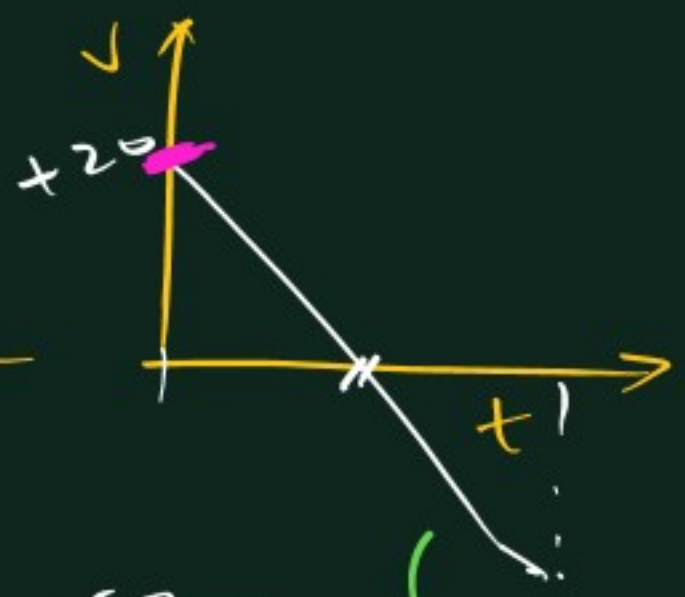
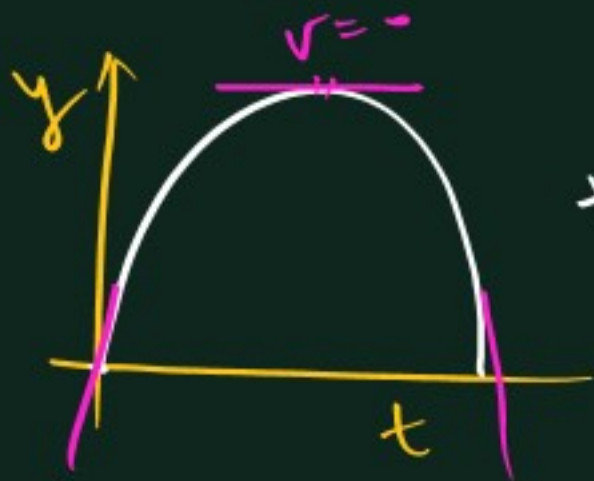
(5s) Area = $v_2 - v_1$
 $\Rightarrow 50 = v_2 - 10$
 $\Rightarrow \underline{v_2 = 60 \text{ m/s}}$

(10s) Area = $v_2 - v_1$
 (0-10)
 $\Rightarrow 50 + (-25) = v_2 - 10$
 $\Rightarrow \underline{v_2 = 35 \text{ m/s}}$



($x-t$, $v-t$, $a-t$)





constant $\ominus$ slope (-10)

2nd way

$$u = +20$$

$$a = -10$$

$$v = u + at \Rightarrow v = 20 - 10t$$

$$\text{or } v = -10t + 20$$

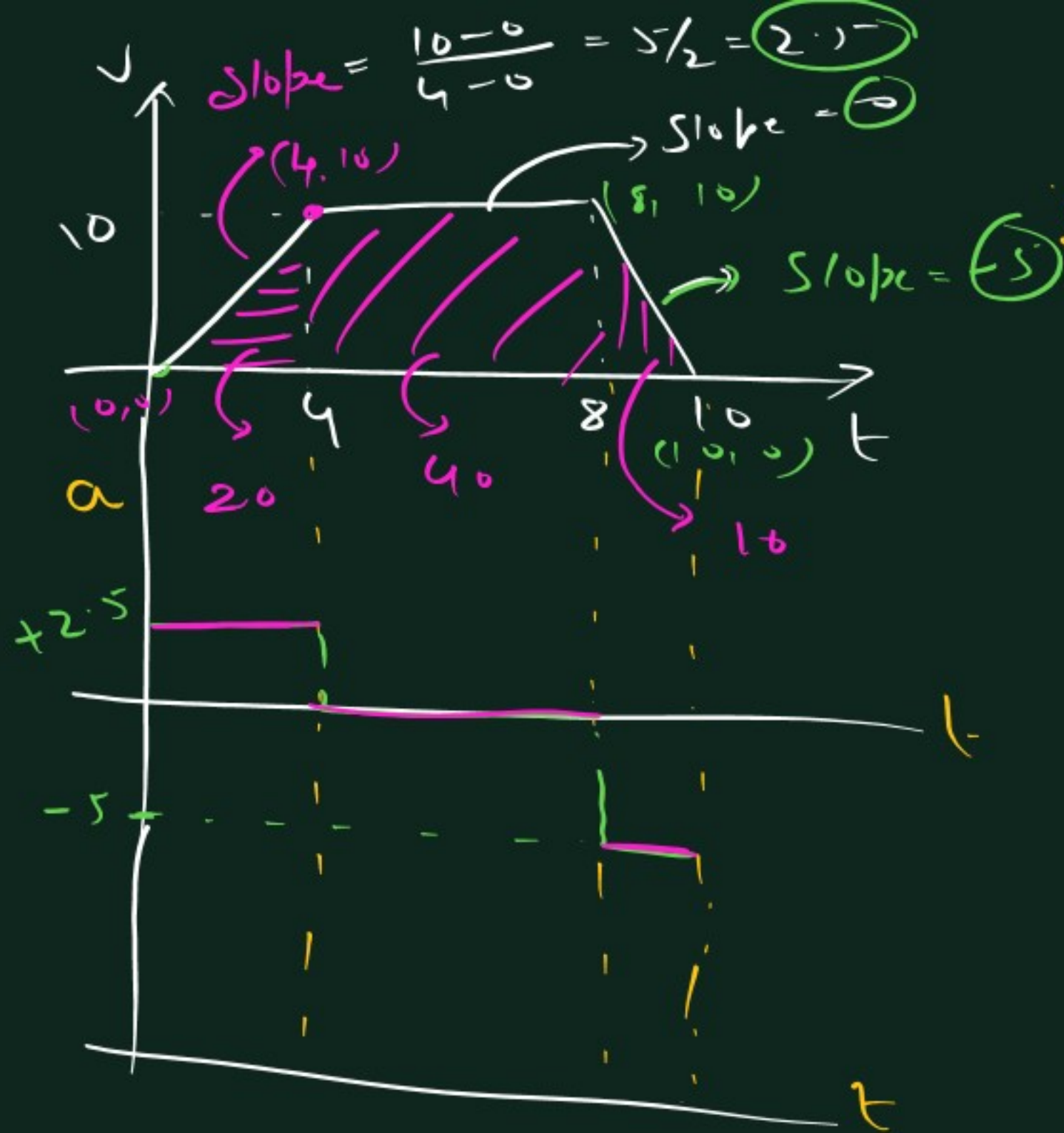
$$(y = -10x + 20) \leftarrow \text{straight line}$$

$$y = ut + \frac{1}{2}at^2$$

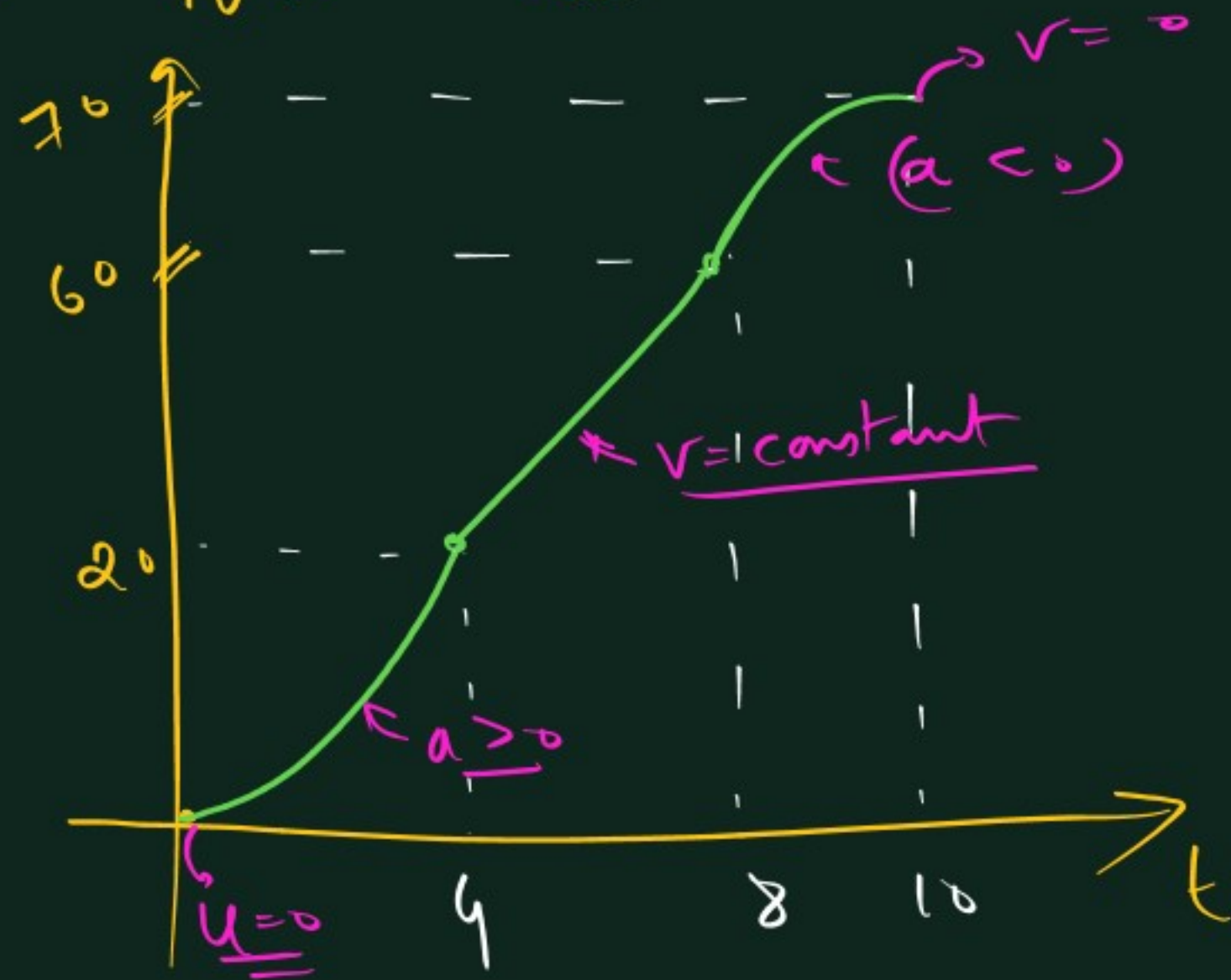
$$= 20t - 5t^2$$

$$y = -5t^2 + 20t$$

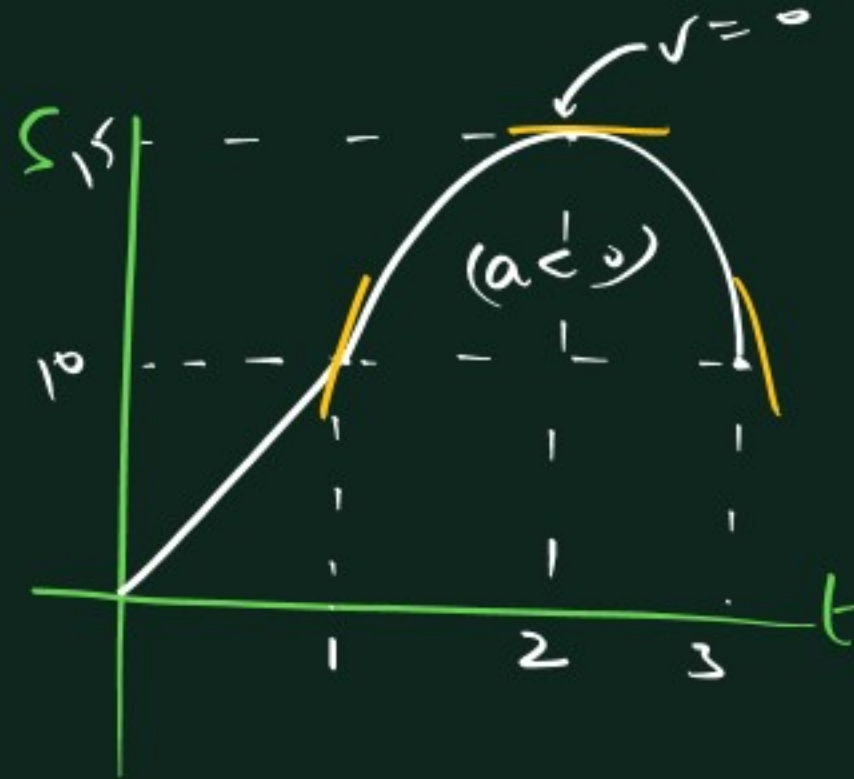
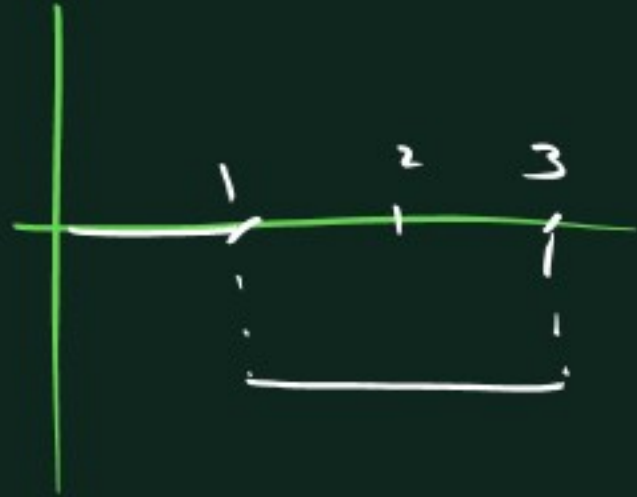
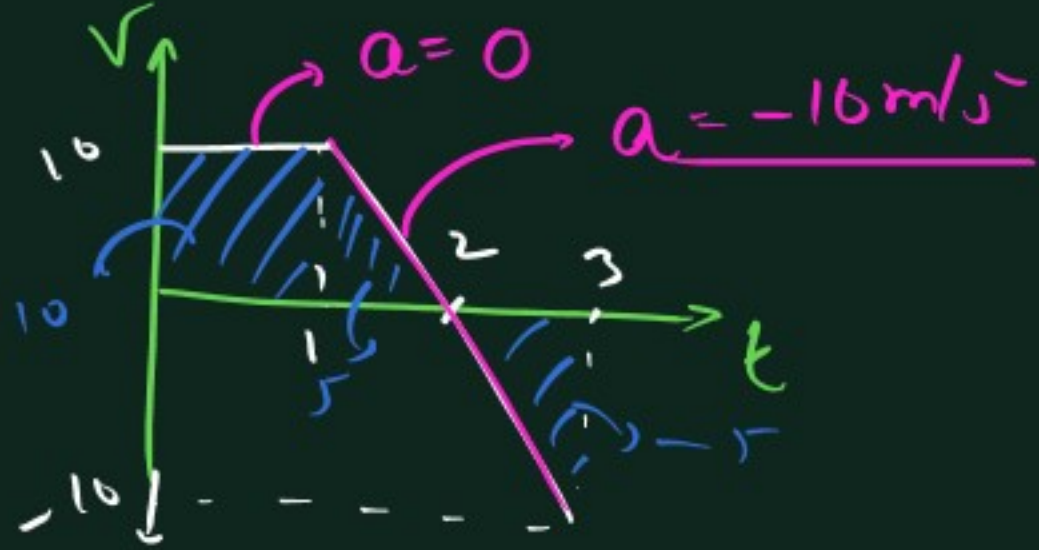




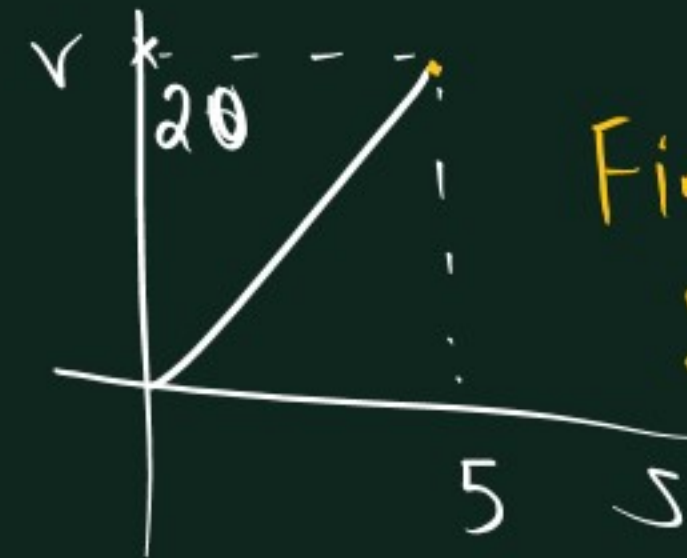
Area $\rightarrow$ S
 Slope $\rightarrow$ a



Q. Draw a-t, s-t



1) $v-s$ & v^2-s

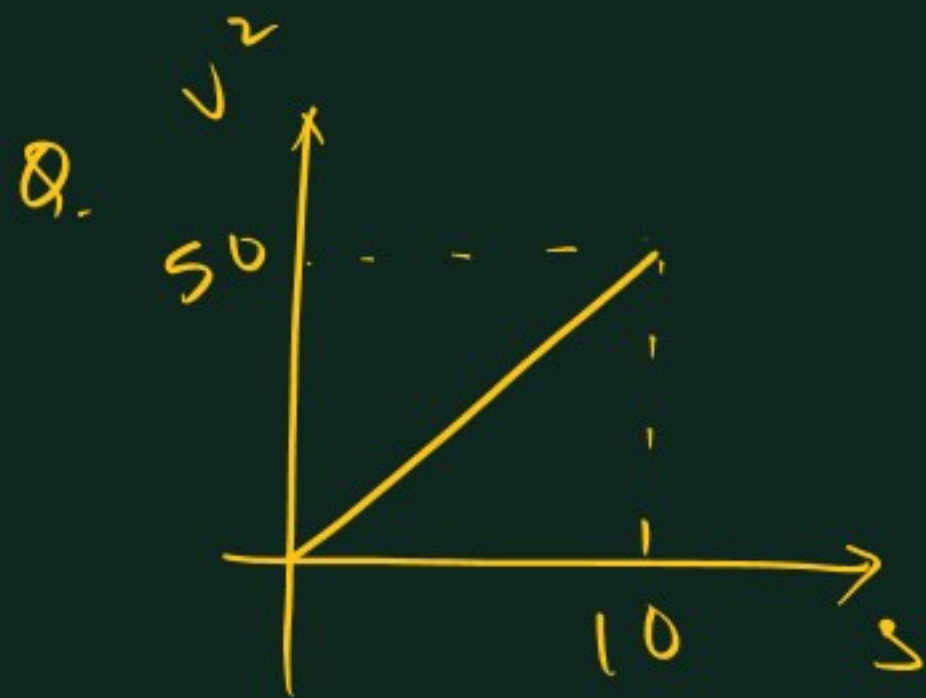


Find 'a' at
 $s = 5m$.

$$a = v \left(\frac{dv}{ds} \right)$$

$$\Rightarrow a = 80 m/s^2$$

Slope = $\frac{20-0}{5-0} = 4$



Find a at $s = 10\text{m}$.

$$\text{Slope} = \frac{d(v^2)}{ds} = \frac{dv^2}{dv} \times \frac{dv}{ds}$$

$$\Rightarrow \text{Slope} = 2v \frac{dv}{ds}$$

$$\text{Slope} = 2 \boxed{v \times \frac{dv}{ds}} \quad a$$

$$\Rightarrow \boxed{a = \frac{\text{Slope}(v^2-s)}{2}}$$

$$\text{Here slope } \frac{50}{10} = 5$$

$$a = \frac{5}{2} = \underline{\underline{2.5 \text{ m/s}^2}}$$

$a-s$

$$a = v \frac{dv}{ds}$$

$$\Rightarrow \int a ds = \int v dv$$

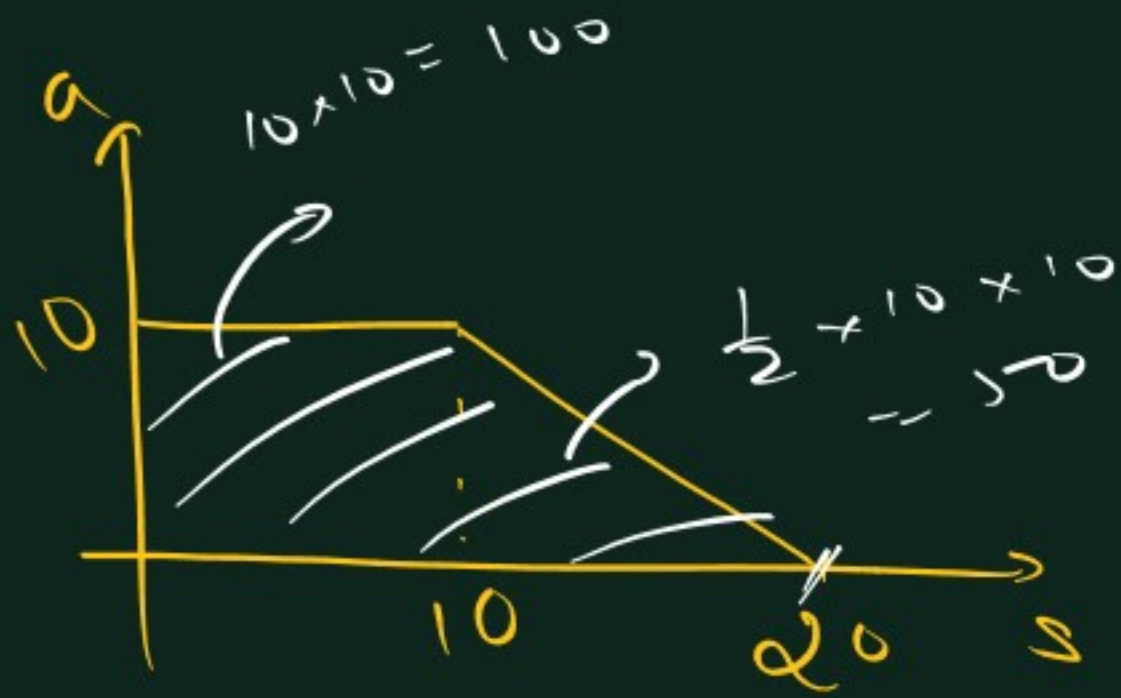
Area under
 $a-s$ graph

$$= \left(\frac{v^2}{2} \right) \Big|_u^v$$

$$= \frac{v^2 - u^2}{2}$$

$$\Rightarrow \text{Area}(a-s) = \frac{v^2 - u^2}{2}$$

$$\Rightarrow \boxed{v = \sqrt{u^2 + 2 \text{Area}(a-s)}}$$

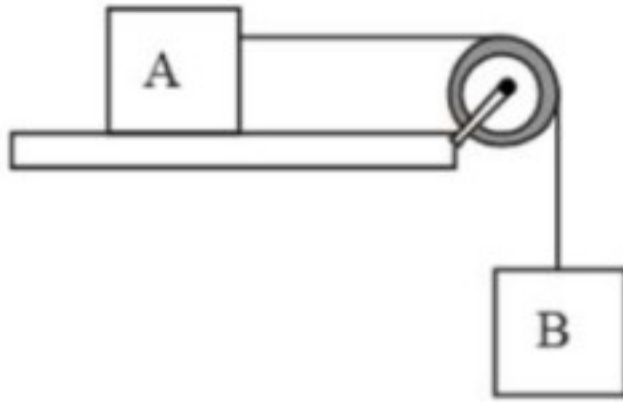


$$v = \sqrt{u^2 + 2(\text{Area})}$$

$$= \sqrt{100 + 2 \times (150)}$$

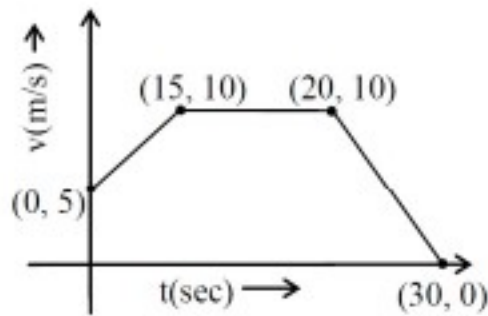
$$= \sqrt{400} = \underline{\underline{20 \text{ m/s}}}$$

- Q.1 In figure coefficient of static friction μ_s between block A of mass 2 kg and the table is 0.2. Maximum mass of block B so that the two blocks do not move, is ($g = 10 \text{ m/s}^2$)



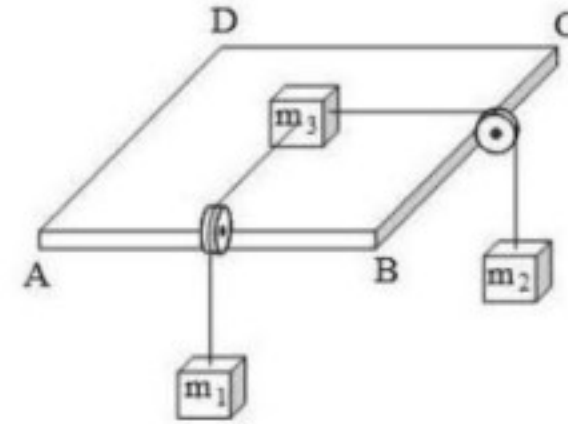
- (A) 2.0 kg (B) 4.0 kg (C) 0.2 kg (D) 0.4 kg

- Q.2 Figure shows velocity-time graph of a particle which moves along a straight line. Then the distance travelled by particle from 0 to 30secs is?



- (A) 175.5 m (B) 200 m (C) 212.5 m (D) 225 m

- Q.3 Three blocks are arranged as shown in which ABCD is a horizontal plane. Strings are massless and both pulleys stand vertical while the strings connecting blocks m_1 and m_2 are also vertical and are perpendicular to faces AB and BC which are mutually perpendicular to each other. If m_1 and m_2 are 3 kg and 4 kg respectively. Coefficient of friction between the block $m_3 = 10 \text{ kg}$ and surface is $\mu = 0.6$ then, frictional force on m_3 is



- (A) 30 N (B) 40 N (C) 50 N (D) 60 N

- Q.4 $\frac{d}{dx}(\tan x)$ is equal to

- (A) $1 - \tan^2 x$ (B) $1 + \tan^2 x$
(C) $1 - \cot^2 x$ (D) $1 + \cot^2 x$

- Q.5 If $y = x^2 + x - 1$ then $\frac{dy}{dx}$ at $x = 1$ is equal to

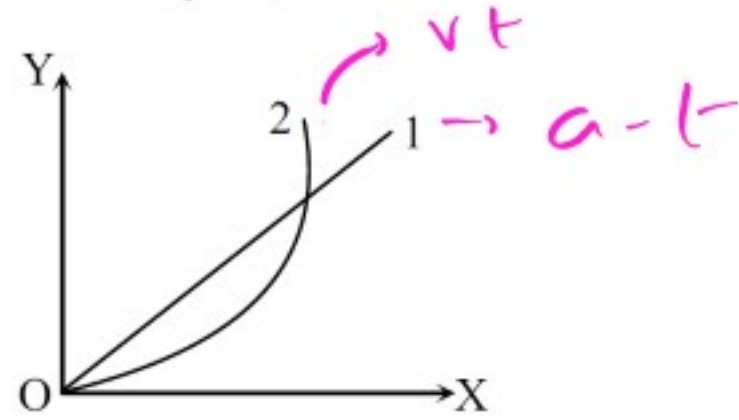
- (A) 3 (B) -3 (C) 0 (D) None of these

- Q.6 $x = at^2$; $y = 2at$, then $\frac{dy}{dx} =$

- (A) t (B) $\frac{1}{t}$ (C) 1 (D) None of these

$$17 \cdot 2$$

- Q.7 A particle is resting over a smooth horizontal floor. At $t = 0$, a horizontal force starts acting on it. Magnitude of the force increases with time according to law $F = \alpha t$, where α is a positive constant. From figure, which of the following statements are correct?



$$a = \frac{F}{m} = \left(\frac{\alpha}{m}\right)t$$

$$\underline{a \propto t}$$

- (A) Curve 1 can be plot of acceleration against time
- (B) Curve 2 can be plot of acceleration against time
- (C) Curve 1 can be plot of velocity against time
- (D) None of the above are correct

$$Q = \int \Rightarrow \checkmark \quad \Rightarrow \quad \int \Rightarrow \checkmark$$

Q.8 The rate of change of velocity gives acceleration. The acceleration of a particle is $a = -\omega^2 x$, where ω is a constant and x is position of the particle at instant t . Find velocity of the particle as function of x , if $v = 0$ at $x = A$.

(A) $v = \omega(A - x)$

~~(B) $v = \omega\sqrt{A^2 - x^2}$~~

(C) $v = \frac{\omega}{2}\sqrt{A^2 - x^2}$

(D) $v = A\omega$

$$v \frac{dv}{dx} = -\omega^2 x$$

$$\Rightarrow \int_0^v v dv = -\omega^2 \int_A^x x dx$$

$$\Rightarrow \left| \frac{v^2}{2} \right|_0^v = -\omega^2 \left| \frac{x^2}{2} \right|_A^x$$

$$\Rightarrow v^2 = -\omega^2(x^2 - A^2) = \omega^2(A^2 - x^2)$$

$$\Rightarrow v = \omega \sqrt{A^2 - x^2}$$

Q.13 A block of mass m is placed on a surface having vertical cross-section given by $y = \frac{x^2}{4}$. If coefficient of friction is 0.5, the maximum height above the ground at which block can be placed without slipping is:

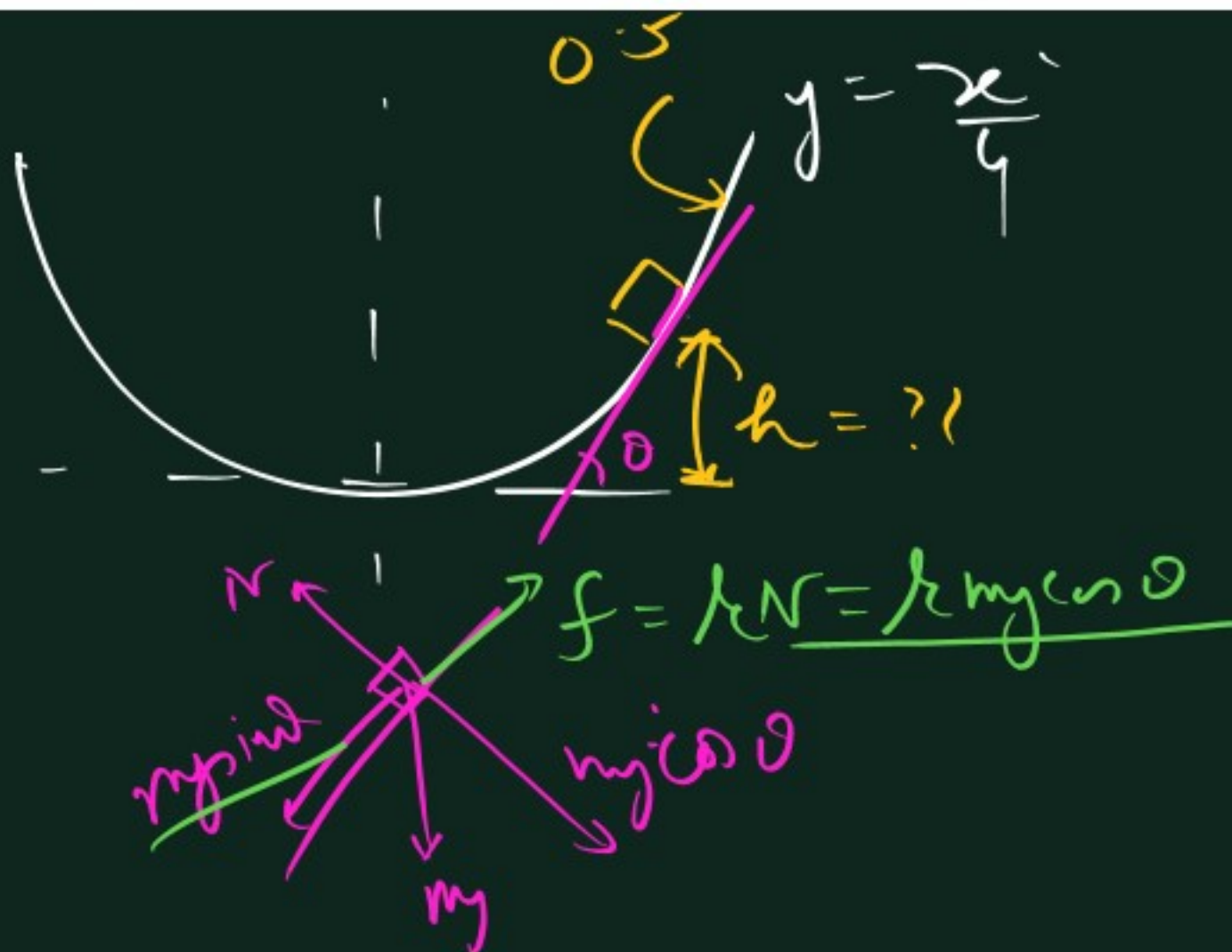
(A) $\frac{1}{2}$ m

(B) $\frac{1}{6}$ m

(C) $\frac{1}{3}$ m

(D) $\frac{1}{4}$ m

$\frac{dy}{dx} = \frac{1}{4} \times 2x = \frac{x}{2}$



$mg \sin \theta = \mu mg \cos \theta$

$\Rightarrow \tan \theta = \mu$

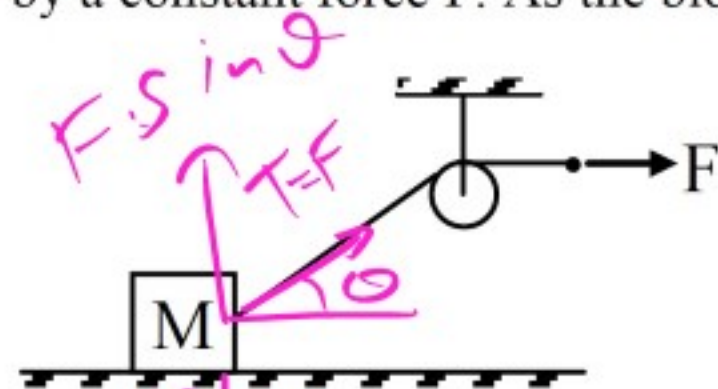
$\frac{dy}{dx}$

$\Rightarrow \frac{x}{2} = \mu$

$\Rightarrow x = 2\mu = 1$

$y = \frac{x^2}{4} = \frac{1}{4}$

Q.14 The block of mass m is placed on a rough horizontal floor and it is pulled by an ideal string as shown by a constant force F . As the block moves towards right the frictional force on block-



(A) remains constant

(B) increases

(C) decreases

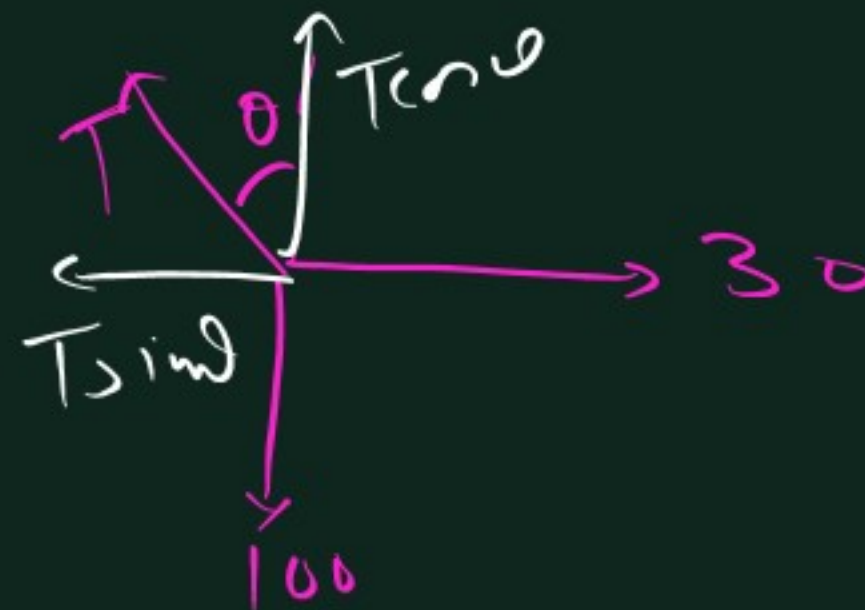
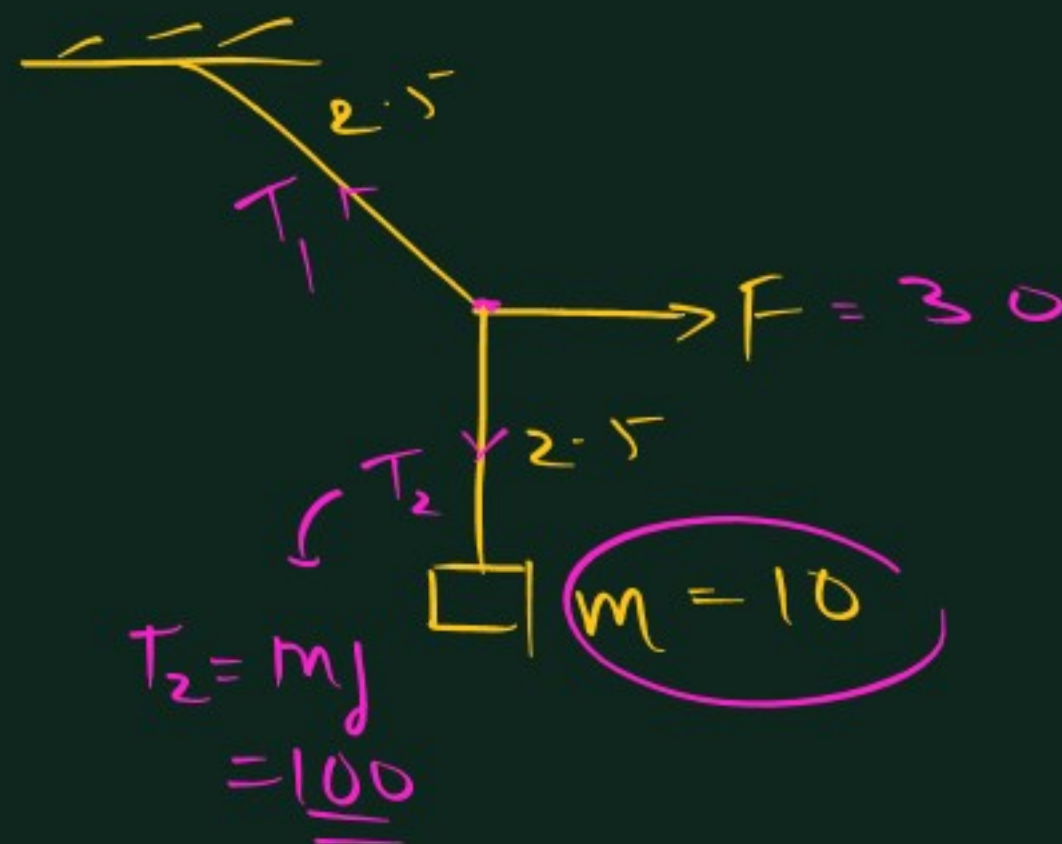
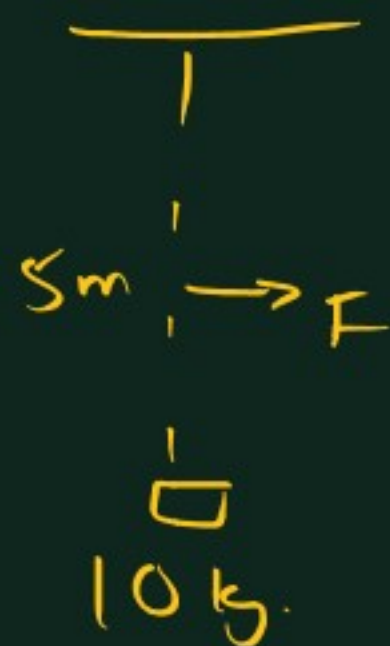
(D) cannot be said

$$N = Mg - F \sin \theta$$

$$\theta \uparrow \Rightarrow \sin \theta \uparrow \Rightarrow N \downarrow$$

$$f_k = \mu N \quad (\downarrow)$$

Q.19 A mass of 10 kg is suspended vertically by a rope of length 5 m from the roof. A force of 30 N is applied at the middle point of rope in horizontal direction. The angle made by upper half of the rope with vertical is $\theta = \tan^{-1} (x \times 10^{-1})$. The value of x is ____ . (Given $g = 10 \text{ m/s}^2$)

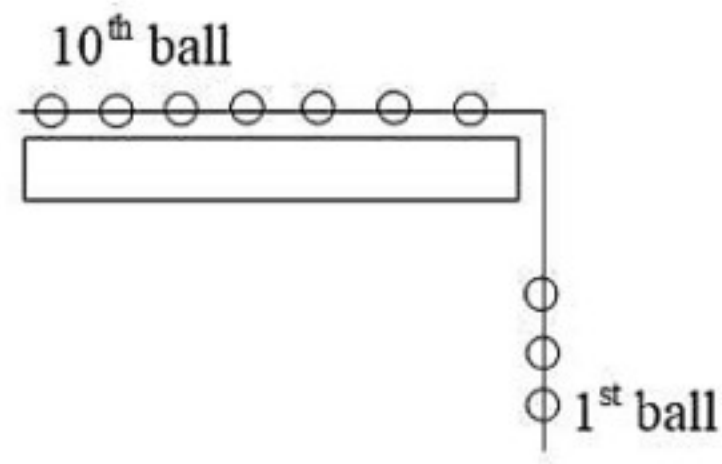


$$T \sin \theta = 30$$

$$T \cos \theta = 100$$

$$\therefore \tan \theta = 0.3$$

Q.22 A system of 10 balls each of mass 2 kg are connected via massless and un-stretchable string. The system is allowed to slip over the edge of a smooth table as shown in figure. Tension on the string between the 7th and 8th ball is _____ N when 6th ball just leaves the table.



$$a = \frac{\text{Net pulling } F}{\text{total mass}} = \frac{6 \cancel{m} g}{10 \cancel{m}} = 0.6g \quad \text{--- (1)}$$



$$T = 3m \times a = 3m \times 0.6g \\ = 3 \times 2 \times 6 = \boxed{36}$$